

Task 1 — Foundation & Environment Setup

1. Cybersecurity Basics

CIA Triad

Confidentiality

Confidentiality means protecting information from unauthorized access. Only users who have the required permission should be able to view or access sensitive data. Examples include passwords, personal information, financial records, and company documents. Authentication, access control, encryption, and permissions are commonly used to maintain confidentiality.

Integrity

Integrity means keeping data accurate, complete, and protected from unauthorized modification. Information should remain unchanged unless an authorized person or system modifies it. Hashing, digital signatures, access controls, and checksums can help maintain data integrity. For example, a file can be hashed before and after transmission to check whether it has been changed.

Availability

Availability means ensuring that systems, applications, networks, and data are accessible when authorized users need them. Hardware failures, network problems, cyberattacks, and system overloads can affect availability. Backups, redundancy, system monitoring, maintenance, and protection against denial-of-service attacks can help maintain availability.

Threat Types

Phishing

Phishing is a social engineering attack in which an attacker attempts to trick users into revealing sensitive information. Attackers commonly use fake emails, messages, websites, or login pages that appear to come from trusted sources. The goal may be to obtain passwords, financial information, or other confidential data.

Malware

Malware is malicious software designed to damage systems, disrupt operations, steal information, or gain unauthorized access. Common types include viruses, worms, trojans, spyware, and ransomware. Malware can enter a system through malicious files, applications, websites, email attachments, or compromised devices.

DDoS

A Distributed Denial-of-Service attack attempts to make a service or system unavailable by overwhelming it with a large amount of traffic or requests. Multiple compromised devices may be used to generate the traffic. The excessive requests consume system or network resources and prevent legitimate users from accessing the service.

SQL Injection

SQL Injection is an attack that targets applications which interact with databases. An attacker attempts to insert malicious SQL commands through application input fields. If the application

does not properly validate or handle input, the injected commands may affect database operations and potentially expose, modify, or delete data.

Brute Force

A brute-force attack repeatedly tries different passwords or authentication combinations until the correct credentials are discovered. The attack can be performed against login systems and other authentication mechanisms. Strong passwords, account lockouts, rate limiting, and multi-factor authentication can reduce the risk of brute-force attacks.

Ransomware

Ransomware is malware that prevents users from accessing their files or systems, commonly by encrypting data. The attacker then demands payment in exchange for restoring access. Regular backups, security updates, endpoint protection, and careful handling of suspicious files can help reduce the impact of ransomware.

Attack Vectors

Social Engineering

Social engineering involves manipulating people into performing actions or revealing information that should remain protected. Instead of directly attacking a technical system, attackers may exploit human trust, fear, curiosity, or urgency. Phishing, impersonation, and deceptive phone calls are common examples.

Wireless Attacks

Wireless attacks target wireless networks and devices. Weak passwords, insecure configurations, outdated protocols, and improperly protected wireless access points can create security risks. Attackers may attempt to intercept traffic, gain unauthorized network access, or exploit weaknesses in wireless communication.

Insider Threats

An insider threat originates from someone who already has authorized access to an organization's systems or information. The person may intentionally misuse their access or accidentally expose sensitive information. Proper access controls, monitoring, authentication, and security awareness can help reduce insider-related risks.

2. Linux Fundamentals

File System Navigation

pwd

The **pwd** command displays the current working directory. It helps identify the exact location inside the Linux file system.

ls

The **ls** command lists files and directories in the current location. Options can be used to display additional information such as permissions, ownership, file size, and hidden files.

cd

The **cd** command is used to move from one directory to another. It allows users to navigate through the Linux file system.

File Permissions

Linux uses permissions to control access to files and directories. Permissions determine whether a user can read, write, or execute a file. The **chmod** command is used to modify permissions, while **chown** is used to change the ownership of files or directories.

Package Management

apt

apt is a package-management command used on Debian-based Linux distributions. It can be used to install, update, remove, and manage software packages.

dpkg

dpkg is a lower-level package-management tool used to install and manage Debian packages. It works with **.deb** package files.

Linux Networking Commands

ifconfig

ifconfig can display and configure network interfaces and their IP-related information.

ping

The **ping** command checks whether another device or host can be reached over a network. It sends network requests and waits for responses.

netstat

netstat provides information about network connections, listening ports, and network-related statistics.

traceroute

traceroute shows the path packets take through different network devices between the source and destination. It can help identify routing paths and possible network problems.

3. Networking Basics

OSI Model

The OSI model divides network communication into seven layers:

1. **Physical** – Transmits raw bits through physical media.
2. **Data Link** – Handles frames and communication between devices on the same network.
3. **Network** – Handles logical addressing and routing using IP.
4. **Transport** – Provides end-to-end communication using protocols such as TCP and UDP.
5. **Session** – Establishes, manages, and terminates communication sessions.
6. **Presentation** – Handles data representation, encryption, and compression.
7. **Application** – Provides network services used by applications.

TCP/IP Model

The TCP/IP model is commonly used to describe how communication takes place across networks. It includes the Network Access, Internet, Transport, and Application layers. Protocols such as IP, TCP, UDP, DNS, HTTP, and HTTPS operate within this model.

DNS

The Domain Name System converts domain names into IP addresses. For example, when a user enters a website domain, DNS helps find the IP address associated with that domain so the system can communicate with the correct server.

HTTP and HTTPS

HTTP is a protocol used for communication between web clients and servers. HTTPS is the secure version of HTTP and uses encryption through TLS to protect communication between the client and server.

IP Addressing

An IP address identifies a device or network interface on an IP network. IPv4 addresses use 32 bits and are normally written in dotted-decimal form. IPv6 uses 128 bits and provides a much larger address space.

Subnetting

Subnetting divides a larger IP network into smaller networks called subnets. It helps organize networks, manage IP addresses, and control network communication.

NAT

Network Address Translation allows private IP addresses to communicate with external networks by translating private addresses into public addresses. NAT is commonly used in routers and helps conserve IPv4 addresses.

4. Cryptography Basics

Symmetric Encryption

Symmetric encryption uses the same secret key for encryption and decryption. The sender encrypts the information using the key, and the receiver uses the same key to decrypt it. The key must be protected because anyone who obtains it may be able to decrypt the information.

Asymmetric Encryption

Asymmetric encryption uses two related keys: a public key and a private key. The public key can be shared, while the private key must remain protected. The two-key approach is used in areas such as secure communication, authentication, and digital certificates.

Hashing

Hashing converts input data into a fixed-length value called a hash. Hashing is generally designed to be one-way, meaning the original input should not be directly recoverable from the hash. Hashes can be used to verify data integrity.

MD5 and SHA

MD5 and SHA are examples of hashing algorithms. MD5 produces a 128-bit hash and is considered unsuitable for modern security-sensitive applications. SHA is a family of hashing algorithms that includes stronger options such as SHA-256.

Digital Certificates

Digital certificates help establish trust between systems and verify the identity associated with a public key. Certificates are commonly used with HTTPS and TLS to support secure communication.

SSL/TLS

TLS provides encryption and security for communication over networks. HTTPS uses TLS to protect web traffic between a client and a server. It helps provide confidentiality, integrity, and authentication.

OpenSSL

OpenSSL is a widely used toolkit for cryptographic operations and TLS-related tasks. It can be used to generate keys, create certificates, calculate hashes, inspect certificates, and perform other cryptographic operations.

5. Tool Familiarization

Wireshark

Wireshark is a network protocol analyzer used to capture and inspect network traffic. It allows users to examine packets and understand protocols, addresses, ports, and communication between systems. Filters can be used to focus on specific types of traffic.

Nmap

Nmap is a network scanning tool used to discover hosts and services on a network. It can identify open ports, running services, and other information about network systems.

Common options include:

- **-sS** – TCP SYN scan
- **-sU** – UDP scan
- **-sV** – Service and version detection
- **-O** – Operating-system detection
- **-A** – Enables several advanced detection features

Burp Suite

Burp Suite is a web application security testing platform. It can intercept and inspect HTTP/HTTPS requests and responses between a browser and a web application. It is useful for analyzing application behavior and testing web security.

Netcat

Netcat is a network utility that can establish TCP or UDP connections. It can be used for network communication, testing ports, transferring data, and troubleshooting network connectivity.